Tab 1

Making Education More Accessible & Reliable : AI–Driven Facilitator–Based Teaching

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Abstract

Facilitator-based delivery can widen access to education by separating expert lesson design from classroom delivery. However, it relies on facilitators, often less experienced than qualified teachers, noticing which learners are struggling. This project designed and evaluated a facilitator decision-support prototype for Year 3–6 reading comprehension. Learners read a short passage and answer two literal and two inferential questions, judging after each answer whether they got it right. The system records accuracy by question type, these self-judgements, hint timing, revisions and timing. Rules grounded in reading-comprehension research turn these signals into a status (“needs support” or “worth a check-in”) shown with the evidence behind it.

With no real classroom data available, the rules were evaluated on 200 simulated classrooms of 30 learners built from five reader profiles, and compared with an accuracy-only threshold and two Random Forest models. On a single passage the refined rules flagged 60.9% of at-risk learners (95% CI 57.9–64.1%), against 56.7% for the score alone, at the same 10% false-alarm rate. They also flagged more of the hard-to-spot poor comprehenders (50.1% vs 42.2%). A Random Forest trained on rule-generated labels only reproduced the rules (45.7% recall). A Random Forest trained on the true simulated labels reached 77.1%, which indicates the value that real outcome labels could add. Pooling three passages raised the rules’ precision to 93.9%, but a threshold-granularity issue reduced their sensitivity to poor comprehenders.

The results show that evidence-linked rules are feasible and modestly better than a score in simulation. The persona parameters are assumptions, though, so validation with real learners and facilitators is the essential next step.

1. INTRODUCTION

Access to quality education remains a significant global challenge, particularly in underserved and resource-limited environments. Many communities continue to face limited access to qualified educators, inconsistent delivery of learning, and barriers to establishing sustainable educational programmes. These challenges affect not only access to education but also the quality and reliability of learning outcomes.

In reply to these challenges, the present project investigates the design of a facilitator-based education platform to strengthen accessibility, affordability, and reliability in educational delivery. The core concept involves separating educational expertise from classroom delivery: lessons are designed by subject-matter experts and delivered by facilitators with structured, step-by-step guidance. This model aims to reduce dependence on highly trained teachers while assuring consistent lesson delivery.

Artificial intelligence (AI) is incorporated into this model as a supporting tool to enhance content generation, personalisation, and resource creation. The primary focus of the project is not on AI in isolation, but rather on the effective integration of AI within structured learning systems that support younger learners.

This report outlines the project scope and objectives and provides a literature review examining how current AI-based and traditional education systems address these challenges, as well as how the proposed model addresses the identified gaps.

2. PROJECT SCOPE & RESEARCH OBJECTIVES

2.1 Problem Statement

Despite advances in educational technology, significant problems persist in delivering consistent, high-quality education at scale. Current systems remain heavily dependent on qualified teachers for both lesson design and delivery, creating a structural bottleneck that limits scalability and contributes to uneven learning quality, particularly in underserved or resource-constrained regions. While AI-supported and online platforms offer greater flexibility, they typically presuppose a level of learner independence that is unsuitable for younger students, who require structured guidance, supervision, and interactive support. Furthermore, access to AI-enhanced content does not guarantee consistent or effective instructional delivery; variability in teaching approaches, the absence of lesson structures, and limited contextual adaptability continue to produce inequitable learning outcomes. These compounding limitations point to a clear gap in existing provision: the absence of a scalable system that decouples expert lesson design from classroom delivery and embeds structured support throughout the learning process.

2.2 Research Aims and Question

The literature in Section 3 shows that AI tools for K–12 learners assume a self-direction younger students often lack, and that facilitator-led delivery depends on the facilitator noticing who is struggling. This project therefore asks whether a lightweight system can turn the interaction data from a short reading activity into evidence-linked recommendations that help a less-experienced facilitator decide which learners need support.

The central research question is:

> **Can a software system convert learner interaction data into evidence-linked recommendations for facilitator decision-making?**

Three sub-questions follow from it:

1. **RQ1.** Do rules grounded in reading-comprehension research flag at-risk learners more reliably than a score threshold, and at what cost in false alarms?

2. **RQ2.** Can these rules detect *poor comprehenders* (fluent readers with weak inference who overestimate their own answers), a group the literature shows teachers tend to miss?
3. **RQ3.** Does a machine-learning model (Random Forest) improve on the rules, given that no real classroom outcome labels are available?

The scope is a proof of concept. The questions are answered on simulated classrooms, not real learners (Section 4.5), so the answers describe how the design behaves under stated assumptions rather than how it will perform in a school.

2.3 General Approach

This project employs a research-informed design methodology that combines literature analysis with system design principles. The study begins with a comprehensive review of research on artificial intelligence in education and current educational delivery models, focusing on their benefits, limitations, and relevance to K–12 contexts. Insights from the literature are synthesised to identify key gaps, particularly regarding structured learning, accessibility, and scalability. Based on these findings, a conceptual facilitator-based platform is proposed to separate lesson creation from classroom delivery while integrating AI as a supporting tool for content generation and learning enhancement. The proposed system is evaluated conceptually by assessing its alignment with the identified gaps and its potential to improve consistency, accessibility, and overall learning effectiveness.

2.4 Objectives

The project aims to:

- Investigate challenges related to access, quality, and teacher dependency in education.
- Analyse current AI-based and traditional learning systems
- Identify limitations in self-directed and technology-supported learning.
- Design a platform model that separates lesson design from classroom delivery.
- Develop a structured facilitator-based learning approach.
- Incorporate AI features to support content generation and learning.
- Demonstrate the feasibility of the proposed system design.
- Evaluate the solution against the gaps currently identified in K-12 education.

2.5 Project Scope

The scope of this project is restricted to the design and development of a proof-of-concept platform for facilitator-based lesson delivery. The primary focus is on establishing the essential system framework and capabilities required to support guided classroom instruction, rather than developing a comprehensive or fully deployable educational system.

2.6 Value of the Project

This project addresses key limitations present in both traditional and AI-supported education systems.

By combining structured lesson delivery with facilitator-based teaching and AI support, the platform aims to:

- Improve consistency in educational delivery.

- Reduce dependence on highly trained teachers.
- Increase accessibility to quality education.
- Enable scalable learning environments.

The proposed system provides a balanced approach by integrating technology with guided instruction, thereby supporting more effective learning experiences for K–12 students.

3. LITERATURE REVIEW

3.1. Overview of Artificial Intelligence in Modern Education

The rapid expansion of artificial intelligence (AI) in education is closely associated with the proliferation of digital and online learning environments. As online platforms became more prevalent, educational delivery increasingly relied on technology, with AI tools integrated to support content delivery, interaction, and assessment *(Kasneci et al., 2023; Alshahrani, 2023).* This shift represents *a greater movement* away from traditional classroom-based instruction toward technology-driven, flexible learning environments, enabling students to engage with digital platforms independently, manage their learning pace, and access resources without continuous teacher supervision *(Lee & Zhai, 2024)*. Concurrently, the educator's role transitioned from direct instruction to facilitation and support *(Touretzky et al., 2019), fostering student-centred learning* with greater emphasis on autonomy and personalised experiences.

Figure 3.1 illustrates that artificial intelligence in education can be conceptualised in three progressive stages: function, utilisation, and impact. This framework illustrates how AI develops from basic technical capabilities to ultimately influencing learning outcomes (Lang, 2025), providing a foundation for examining specific methods of AI integration within educational *environments*.

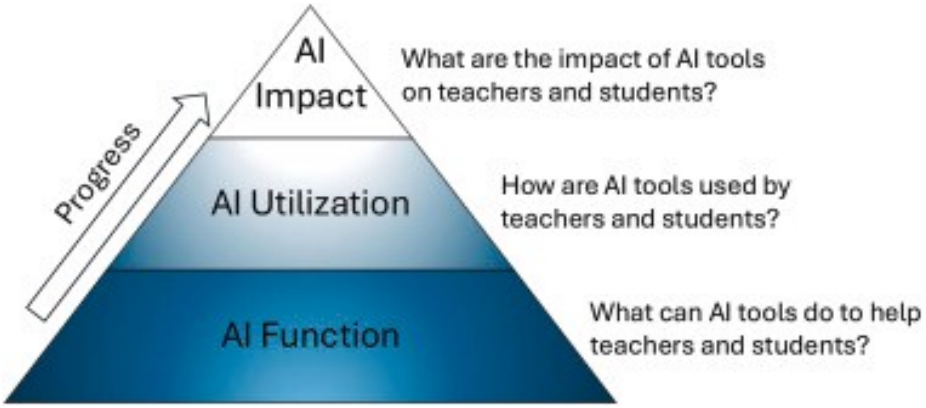


Figure 3.1: Progressive Stages of Artificial Intelligence Application in Education (sourced from Lang, 2025)

Despite these advancements, the transition to AI-enabled platforms has not benefited all learners equally. While such platforms have improved educational access, they have also introduced variability in learning experiences. Some students adapted effectively to independent environments, whereas others struggled without structured guidance *(Vilcarino & Langreo, 2025)*. Additionally, the rapid adoption of AI has surpassed the development of effective implementation frameworks, resulting in inconsistent application across educational contexts (Lang, 2025). This tension between flexibility and consistency, along with the absence of standardised guidance, illustrates the need to better understand how AI supports learning, particularly for younger students who require structured and guided environments.

3.2. Applications and Benefits of AI in Younger Education (K-12)

AI has been implemented in education through personalisation of learning, task automation, and support for teaching processes. A prominent application is adaptive learning systems, which adjust content based on student performance, allowing learners to access materials tailored to their needs and progress at an individual pace *(Lang, 2025; Pardamean et al., 2022)*. For example, a study with 269 primary school students demonstrated that AI-based personalised learning significantly improved student performance, showing the effectiveness of adaptive approaches *(Pardamean et al., 2022)*. Furthermore, AI systems deliver immediate feedback, enabling students to identify errors and enhance understanding in real time *(Samala et al., 2024)*. This continuous feedback loop promotes active learning and facilitates rapid error correction, particularly in practice-intensive subjects such as mathematics and language learning, thereby increasing efficiency compared to traditional methods where feedback is delayed.

AI has also contributed to enhanced accessibility and efficiency in education. AI-based platforms provide students with the ability to access learning materials at any time and from any location, increasing flexibility and inclusivity, especially for those in remote or underserved regions where traditional resources are scarce (Rane, 2024). Additionally, AI adapts content to accommodate various learning styles and abilities, and automates administrative tasks such as grading, content generation, and data analysis *(Pisica et al., 2023)*. These efficiencies reduce educators' administrative workload, enabling greater focus on student interaction and support, and promote scalability and resource optimisation in educational settings.

Despite these advantages, research on AI's effectiveness in education has produced mixed results. An umbrella review of 102 studies indicated that although AI enhanced engagement and personalisation, its success was highly dependent on implementation quality and student interaction *(Lang, 2025)*. While some studies reported notable improvements in learning outcomes, others observed minimal impact, indicating that AI alone does not ensure improved learning and must be thoughtfully integrated into educational processes. This issue is particularly evident in the context of self-directed learning, where AI's effectiveness is closely linked to learners' skills and dispositions.

3.3. AI in Self Directed Learning (SDL)

Self-directed learning (SDL) is an educational approach in which students assume responsibility for planning, monitoring, and evaluating their own learning. Research indicates that AI tools can facilitate SDL by offering personalised guidance, feedback, and adaptive learning pathways, *thereby encouraging* students to take a more active role and enhancing motivation and engagement (Ng et al., 2024; Achuthan, 2025). A meta-analysis of 32 studies with 3,029 participants found that AI-based interventions significantly improved both self-regulated and self-directed learning *(Achuthan, 2025). Similarly, a study involving 74 students over three weeks reported improvements in* engagement, motivation, and learning outcomes *(Ng et al., 2024)*.

As illustrated in Figure 3.2, AI interventions support different dimensions of self-regulated learning, including cognitive, motivational, and behavioural processes, which contribute to the development of self-directed learning *(Achuthan, 2025)*.

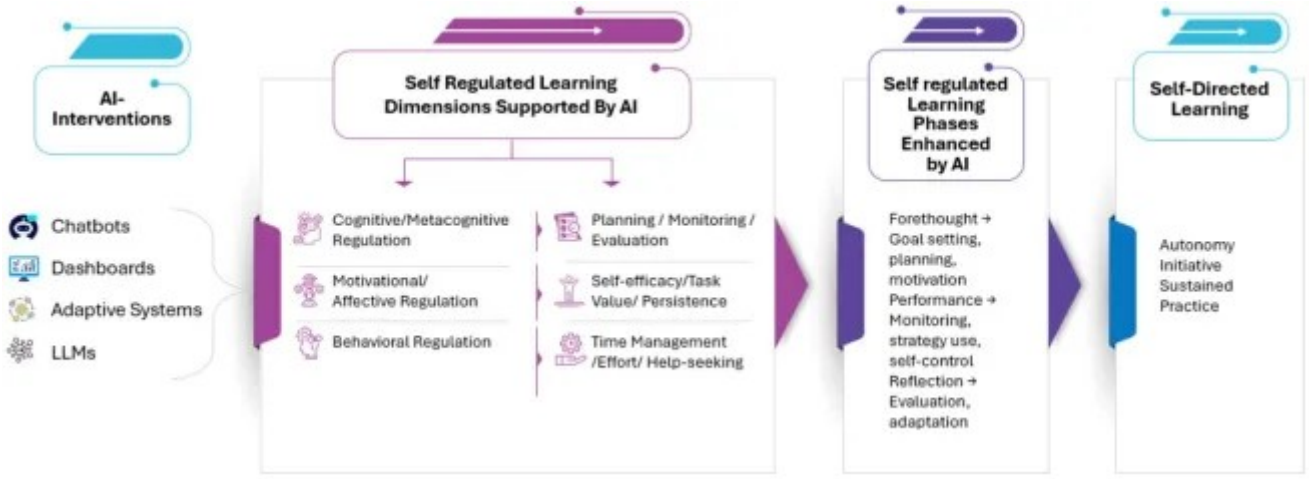


Figure 3.2: AI–Regulation–Autonomy Pathway in Learning (sourced from Achuthan, 2025)

This demonstrates AI's function as a scaffold that assists learners in planning, monitoring, and evaluating their progress. However, its effectiveness is contingent upon the learner's ability to engage independently with these processes. Research consistently shows that students need strong self-regulation skills, such as time management, discipline, and progress monitoring, to use AI tools effectively (Shen & Teng, 2024). Earlier studies also found that students in online environments often struggled with independent learning *(Chen, 2009; Kim et al., 2014). Without these foundational skills, students not only failed to benefit from AI's features but also risked developing over-reliance, which is associated with diminished critical thinking and problem-solving abilities (Kasneci et al., 2023).*

Collectively, these findings suggest that AI is most effective when learners possess strong self-directed learning skills, a condition less common among younger students in K–12 settings. This raises significant concerns regarding the suitability of AI-driven approaches for this age group, which are examined in the subsequent section.

3.4. Limitations of AI for Younger Learners (K-12)

Despite its potential, AI exhibits several limitations when implemented with younger learners in K–12 education. A primary challenge is the presumption of self-directed learning; younger students frequently struggle to manage their learning independently, resulting in reduced engagement and inconsistent outcomes *(Chen, 2009)*. Numerous studies have found that students often rely excessively on AI tools, using them to complete tasks without fully understanding the material and, in some cases, bypassing essential learning processes (NPR, 2026). A review of 22 studies identified additional risks, such as cognitive dependency, diminished creativity, and negative effects on student well-being (Tao et al., 2026), *raising broader concerns that* AI may supplant rather than support learning.

Researchers have also observed that AI systems are unable to provide the structured guidance and meaningful human interaction essential for younger learners *(Healey, 2025)*. Unlike teachers, AI cannot adequately address students' emotional and behavioural needs, which may decrease engagement and motivation, particularly among younger students. Additional studies have identified psychological and developmental concerns, such as reduced confidence, social isolation, and increased dependence on technology *(Tao et al., 2026), which are particularly significant for learners still developing foundational cognitive and social skills. Broader systemic issues*, including data privacy, ethical considerations, and unequal access to technology, further contribute to inconsistent implementation and variable learning experiences across educational contexts (Lang, 2025).

In summary, these findings demonstrate that although AI presents potential benefits in K–12 education, it does not fully address the developmental and pedagogical requirements of younger learners.

3.5. Implications for AI-Supported Learning System Design

The literature suggests that effective integration of AI in education necessitates careful system design that considers both its advantages and limitations. Researchers consistently emphasise that AI should serve as

a complement to, rather than a replacement for, teachers, supporting the delivery of structured and meaningful learning experiences *(Kasneci et al., 2023)*.

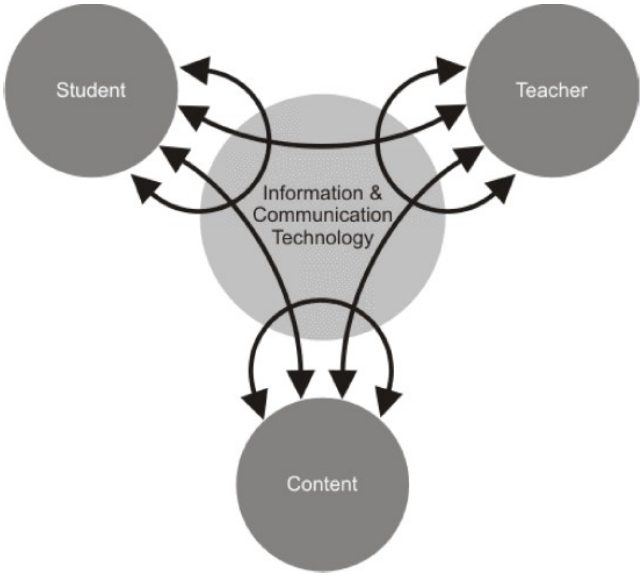


Figure 3.3: Interaction Between Students, Teachers, and Learning Content in AI-Supported Learning Environments (sourced from Pardamean et al., 2022)

Figure 3.3 demonstrates that AI systems facilitate interactions among students, teachers, and learning content, fostering more connected and adaptive learning environments (Pardamean et al., 2022). This underscores the importance of integrating AI within structured systems that guide students through learning processes. Research highlights the value of guided learning environments, where students follow clear steps and receive consistent support; structured lesson flows help maintain engagement and minimise confusion or disconnection (Lang, 2025). Rather than relying solely on AI, educational systems should enable facilitators to guide students, monitor progress, and provide additional support as needed. This hybrid approach combines the scalability of AI with the effectiveness of human interaction (Pisica et al., 2023). AI should be used selectively to enhance, rather than replace, core learning processes *(Samala et al., 2024)*, *supporting* activities such as content generation, feedback, and practice, while educators remain responsible for ensuring meaningful engagement. Centralised content management and structured lesson design further ensure consistent educational quality for all students, regardless of location or facilitator.

Overall, the literature suggests that future educational systems should incorporate AI within structured, facilitator-supported environments, balancing the scalability and flexibility of AI with the consistency and human connection necessary for effective learning.

3.6 Identified Gaps in AI-Supported Education

Building on the preceding sections, the literature reveals several consistent gaps in the current implementation of AI within K–12 education. While AI demonstrates significant potential to enhance personalisation, accessibility, and learning efficiency, existing systems do not consistently address the pedagogical and developmental needs of younger learners, nor do they adequately account for the practical constraints of real-world delivery at scale. These gaps inform the need for more structured, scalable, and context-sensitive approaches to AI-supported education.

3.6.1 Personalisation and the Delivery Gap

A substantial body of research confirms that personalised learning improves student engagement, achievement, and motivation by adapting content to individual needs and abilities *(Pardamean et al., 2022; Samala et al., 2024)*. AI systems have demonstrated effectiveness in supporting this through adaptive content delivery and real-time feedback. However, the literature also reveals that current implementations focus predominantly on content-level personalisation, with considerably less attention given to how that learning is structured and delivered in practice. While materials may be tailored to individual students, the overall learning experience often remains inconsistent and context-dependent, pointing to a gap between

personalised content generation and structured instructional delivery. Effective AI-supported learning, therefore, requires integrating both elements, not merely adapting content.

3.6.2 Dependence on Teacher Expertise and Scalability Constraints

The literature consistently highlights that current educational models remain heavily dependent on teacher expertise for both the design and delivery of learning experiences. Although AI can support content generation and feedback, its effective implementation still relies on educators to structure and facilitate learning *(Lang, 2025)*. This reliance creates significant scalability challenges, particularly in contexts where access to highly trained educators is limited or unreliable. The findings suggest that instructional design and classroom delivery are not sufficiently separated in existing systems, a bottleneck that constrains the reach and consistency of AI-supported education and that the literature does not yet offer a clear solution to.

3.6.3 Lack of Structured Guidance in AI-Supported Learning

A consistent limitation identified across studies is the assumption of self-directed learning capability in AI-based environments. As established in sections 3.3 and 3.4, younger learners frequently lack the self-regulation skills required to engage with AI tools independently and effectively *(Chen, 2009; Shen & Teng, 2024)*. Without a clear instructional structure, students are prone to confusion, disengagement, or over-reliance on AI-generated outputs, outcomes that are particularly damaging in K–12 contexts where foundational skills are still being developed. The literature, therefore identifies a gap in which flexibility is prioritised over structured support, thereby reducing the effectiveness of AI for learners who arguably need it most.

3.6.4 Accessibility and Equity Gap

Although AI has been shown to improve access to learning materials through flexible, on-demand delivery *(Rane, 2024)*, access to high-quality education remains uneven. Factors such as teacher availability, institutional infrastructure, and geographic location continue to shape educational outcomes in ways that AI alone cannot resolve *(Lang, 2025)*. Current implementations primarily enhance access to content, rather than addressing the broader challenge of delivering consistent, high-quality learning experiences across diverse contexts. This distinction between access to resources and access to effective education represents a meaningful gap in the literature, one that system design must explicitly address.

3.6.5 Limited Adaptability Across Contexts

While AI systems can personalise content at the individual level, the literature indicates that many platforms lack adaptability across broader educational contexts. Variations in curriculum, learner age, subject area, and cultural environment are not consistently accommodated, limiting the transferability of AI-supported systems across settings (Achuthan, 2025). Most studies evaluate AI tools within specific institutional or geographic contexts, making it difficult to draw generalisable conclusions about their effectiveness elsewhere. The findings, therefore, highlight a need for systems that support flexible instructional design at the platform level, enabling content and delivery to be modified to meet diverse contextual requirements rather than being fixed to a single educational environment.

3.6.6 Synthesis

Collectively, the literature reveals a consistent pattern: current AI-based educational systems emphasise content personalisation but overlook instructional delivery; remain dependent on teacher expertise in ways that limit scalability; assume levels of learner independence that are often inappropriate for younger students; improve access to content without ensuring equitable learning outcomes; and lack the adaptability required to function effectively across diverse educational contexts. Taken together, these gaps indicate that the promise of AI in education has yet to be fully realised in practice, particularly for K–12 learners in non-traditional or under-resourced settings. Addressing these limitations is essential to achieving the broader goals of accessibility, consistency, and quality in education and provides the foundation for the system design proposed in this study.

4. METHODOLOGY

4.1 Design Approach

- Primary approach: an interpretable machine learning model classifies learner interaction sessions into behavioural categories, replacing hand-authored rule conditions as the main recommendation-generation mechanism.
    - Interpretable model family chosen deliberately (e.g. decision tree / rule-extraction algorithm) rather than a black-box model (e.g. neural network) — preserves the evidence-linked, traceable property central to the framework's value proposition.
- Rule-based engine retained in two supporting roles: (1) generating initial training labels in the absence of real classroom outcome data, and (2) serving as an interpretability/comparison baseline in evaluation.
- [Update your research question wording if needed — e.g. from 'can interaction data be converted into evidence-linked recommendations' to explicitly reference an ML-based classification approach, while keeping the evidence-linked requirement intact.]

4.2 Data Model

- Strongly-typed TypeScript data structures used throughout the pipeline (capture → feature extraction → model → dashboard).
    - [List core types: LearnerResponse, InteractionEvent, FeatureVector, Recommendation, and key fields on each.]
- Interaction signals: response timing, self-reported confidence, hint usage, answer revisions[, passage revisits] — now treated as the ML model's input feature vector.

4.3 Data Generation and Labelling Strategy

- No real classroom outcome data exists at this stage of the project, so training data was generated synthetically.
- Learner personas defined with known, intended interaction patterns (e.g. struggling-but-persistent, confident guesser, typical performer, careful reviser) — see Table 4.1.
- The existing rule-based engine was applied to synthetic sessions to generate initial pseudo-labels (behavioural category per session).
- [State dataset size, e.g. N synthetic sessions generated, split into training/test sets, e.g. 80/20.]
- Explicit limitation: labels reflect the logic encoded in the rule engine and persona design, not verified real-world facilitator judgements — carried through honestly into Section 5.5.

4.4 ML Model Design

- Candidate model: [decision tree classifier / rule-extraction algorithm — pick one as primary and justify].
- Input: extracted feature vector per learner session (timing, confidence, hints, revisions[, revisits]).
- Output: predicted behavioural category, mapped to an evidence-linked recommendation in the same format as the original rule-based output.
- Explainability mechanism: [decision path through the tree / feature importance scores] surfaced alongside each prediction so the dashboard can still show facilitators which interaction data drove the recommendation.
- [Optional secondary approach] Unsupervised clustering (e.g. k-means) applied to the same feature vectors to identify behavioural groupings not explicitly hand-coded as rules or personas — positioned as exploratory evidence that ML can surface patterns beyond the original rule set.

4.5 Rule-Based Baseline (Retained)

- Original rule-based engine retained unchanged, now serving as: (a) the source of initial training labels, and (b) a comparison baseline in evaluation.
- [Briefly redescribe rule structure here in 1–2 bullets — condition → indicator → linked evidence — referencing the fuller version already drafted in the previous section, now reframed as baseline rather than primary contribution.]

4.6 Implementation

- Stack: Next.js + TypeScript for the application; [decide: TensorFlow.js for in-browser ML, or a small Python service (scikit-learn) called from the Next.js app — state which and why, e.g. TensorFlow.js keeps everything client-side consistent with current storage approach, while scikit-learn offers more mature interpretable-model tooling].
- Feature extraction logic shared between the rule engine and the ML pipeline to keep both approaches comparable on identical inputs.
- Storage remains browser-based for this iteration, consistent with existing scoping decision (see Future Work for backend migration).

4.7 Testing and Validation Approach

- Train/test split on synthetic dataset (e.g. 80/20) to evaluate generalisation rather than memorisation of training personas.
- [If dataset small, mention cross-validation instead of a single split.]
- Rule-based engine run on the same test set to produce a like-for-like comparison (Section 5.2).

5. EVALUATION

5.1 Evaluation Scope and Rationale

- Evaluates: (1) predictive performance of the ML model against pseudo-labelled synthetic data, (2) explainability/traceability of ML outputs, (3) comparison against the rule-based baseline, (4) technical feasibility of the end-to-end pipeline.
- Explicitly out of scope: validation against real classroom outcomes, facilitator usability, educational effectiveness — Future Work (Section [X]).

5.2 Model Comparison: Rule-Based vs. ML

- Table 5.1 compares the ML model against the retained rule-based baseline on the same synthetic test set.

Aspect	**Rule-Based Baseline**	**ML Model (Decision Tree / Rule-Extraction)**
Accuracy vs. pseudo-labels	[fill in — reference point]	[fill in]
Explainability	Full — hand-authored conditions	Partial — decision path / feature importance
Coverage of behavioural patterns	Limited to hand-authored rules	Can surface patterns not explicitly coded
Dependency	Domain reasoning only	Depends on quality/coverage of training labels

- [Add 2–3 sentences interpreting the table once you have real numbers — e.g. does ML meet/exceed rule-based accuracy? What's the explainability trade-off in practice?]

5.3 End-to-End Traceability Walkthrough

- Retain manual worked examples from the rule-based version, but add: for the same example sessions, show the ML model's prediction, decision path/feature importance, and resulting recommendation — side-by-side with the rule-based output.
- This directly demonstrates the ML approach still produces evidence-linked, traceable recommendations, not just accurate ones.

5.4 Correctness and Robustness Checks

- Boundary/negative case testing carried over from rule-based evaluation, re-run against the ML model.

- [Add: does the ML model behave sensibly on edge cases outside the training distribution? This is a real weak point for ML models trained on small synthetic sets — worth testing and reporting honestly.]

5.5 Limitations of Current Evaluation

- ML model trained and evaluated entirely on synthetic, rule/persona-derived data — performance reflects consistency with the label-generation logic, not validated real-world accuracy.
- Explainability of the ML model is partial compared to the fully transparent rule-based baseline — an explicit trade-off, not a strict improvement.
- No real classroom data, facilitator review, or educational outcome validation — Future Work (Section [X]).
- [Carry over: single literacy activity, single age group — generalisability untested.]

Facilitator Decision-Support Framework

Methodology Update: Machine Learning Development, Reading Comprehension Grounding, and Simulated Classroom Evaluation

1. Machine Learning Development Methodology

1.1 Design Approach

The framework's primary recommendation mechanism is being extended from a purely rule-based system to a Random Forest classifier, in line with the project's AI-driven learning objective. The original rule-based engine is retained in two supporting roles: as the source of initial training labels in the absence of real classroom outcome data, and as an explainable comparison baseline. Random Forest was selected specifically because it supports feature-importance scoring, allowing predictions to remain traceable to the interaction data that produced them, preserving the evidence-linked property that is central to the framework's value proposition, rather than adopting a fully black-box model.

1.2 Reading Comprehension Research Informing the Activity Design

Beyond general learning-analytics literature, the activity design and feature set have been revised based on research specific to reading comprehension in the Year 3-6 age band:

- **Fourth-grade slump / stages of reading development (Chall, Jacobs, & Baldwin, 1990)**: justifies the Year 3-6 scope as targeting the developmental window where comprehension difficulty first becomes distinguishable from decoding difficulty, as texts shift from familiar narratives to more demanding content.
- **Literal vs. inferential comprehension**: research shows these develop on different trajectories (fluency and literal comprehension improve steadily across a school year, while inferential comprehension does not follow the same pattern). Each question in the activity is now tagged as literal or inferential, with accuracy tracked separately for each, rather than blended into one score.
- **Postdictive metacognitive judgment**: a study of 3rd-5th graders found that children with reading comprehension difficulties showed measurably worse metacognitive monitoring than peers, using a postdictive method where learners judge each answer as correct, incorrect, or uncertain immediately after answering. This method has been adopted directly, replacing a single upfront confidence rating, producing a bias score (judged accuracy minus actual accuracy) and an accurate-estimation count.
- **Comprehension monitoring and repair strategies**: comprehension monitoring is understood as detecting a difficulty and then attempting a repair (rereading, seeking help). The activity now tracks whether a hint request or difficulty signal is followed by an observable repair action (a passage reread or an answer revision), rather than only counting these behaviours in isolation.

- **Pre-attempt hint use**: multi-semester research on unproductive hint use found that hints requested before any genuine attempt are a stronger indicator of struggle than raw hint counts. The activity now distinguishes pre-attempt hint requests from hints requested after an attempt has been made.
- **Answer-revision direction**: a meta-analysis of answer-changing behaviour found the majority of answer changes are from incorrect to correct, meaning revisions are, on average, a positive rather than negative sign. Revisions are therefore no longer treated as a struggle indicator in isolation.
- **Facilitator judgement gap**: research comparing teacher judgement to actual reading assessment results found that teachers' assessments of reading difficulty aligned more closely with spelling results than with the reading skills actually in question, directly supporting the project's motivation of providing structured, evidence-linked support to less-experienced facilitators.

1.3 Data Generation and Model Training

As no real classroom outcome data exists at this stage of the project, training and evaluation data are generated synthetically using defined learner personas with interaction patterns grounded in the research above. The rule-based engine is applied to this synthetic data to produce initial pseudo-labels, which are then used to train the Random Forest classifier. This mirrors a recognised approach in educational data mining and AI-in-education research, where synthetic or simulated learner data is used to complement scarce real-world datasets.

2. Evaluation Method: Simulated Classroom

2.1 Why This Method Was Selected

No real classroom outcome data is available at this stage, so there is no ground truth against which to evaluate the framework's accuracy in a real-world setting. A simulated classroom addresses this by generating learner data with a known, controlled ground truth (the persona each synthetic learner was generated from), allowing standard classification metrics, accuracy, precision, recall, and false-positive rate, to be computed directly, rather than relying on a small number of illustrative manual examples.

2.2 Precedent in the Research Literature

This is not a novel or improvised method. “Simulated learners” are an established evaluation technique within Artificial Intelligence in Education (AIED) research, with a lineage spanning roughly three decades:

- Greer and Mark (2016), revisiting their own influential 1993 evaluation framework, explicitly list simulated learners alongside learning curves and kappa scores as a standard, recognised technique for evaluating intelligent tutoring systems and adaptive learning environments.
- Early simulated learners such as Sierra (VanLehn, 1990) and Cascade (VanLehn, 1999) were developed specifically to test tutoring system designs before real deployment, a concept the AIED Society's founder John Self described as an “executable student model” functioning as a “wind tunnel.”
- More recent systems (e.g. SimStudent; Koedinger, Matsuda, MacLellan, & McLaughlin, 2015) and testbeds (TutorGym; Weitekamp, Siddiqui, & MacLellan, 2025) continue this tradition, using simulated or synthetic learner populations to evaluate tutoring and decision-support systems at a scale and level of control not possible with real classroom data alone.
- A systematic review (Käser & Alexandron, 2024) proposes a “Turing-like test” for validating whether simulated learners genuinely resemble real ones, an important, honest limitation acknowledged in this project's evaluation, since the personas used are self-defined rather than derived from a validated real-learner distribution.

2.3 Implementation

The evaluation is implemented as follows:

- Four learner personas are defined (typical performer, struggling but persistent, confident guesser, careful reviser), each with interaction-feature distributions grounded in the research described in Section 1.2.
- Each simulated classroom contains 30 synthetic learners, with personas assigned according to realistic proportions.
- Rather than a single classroom, 20 independent classrooms are generated to ensure results reflect a stable pattern rather than one favourable random draw.
- Both the rule-based engine and the Random Forest model are run on every simulated classroom. Because each learner's persona is known, accuracy, precision, recall, and false-positive rate can be computed for each model against this ground truth and averaged across all classrooms.
- Workload-relevant metrics are also captured, including the class-level triage ratio (how many of the 30 learners are flagged at all) and the information-reduction ratio (raw interaction data points per learner versus what is shown on the facilitator dashboard) these support, but do not by themselves prove, the project's workload-reduction motivation.

2.4 How This Will Be Written Into the Final Report

- Methodology (Section 4): persona design, classroom generation process, and the rationale for repeated simulation, with the AIED simulated-learner literature cited to justify the method itself.

- Evaluation (Section 5): simulated classroom design, rule-based vs. Random Forest model comparison table (accuracy/precision/recall/false-positive rate), workload-relevant metrics, and an end-to-end traceability walkthrough using example simulated learners.
- Limitations (Section 5, closing): explicit acknowledgement that ground truth reflects self-defined design assumptions rather than validated real-world labels, citing Käser and Alexandron's (2024) Turing-like test critique directly, alongside the need for real classroom and facilitator validation as future work.

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1) ML Methodology + Reading Comprehension Research

Methodology pipeline:

- **Design approach**: Random Forest as primary model (predictive + explainable via feature importance); rule-based engine retained as (a) pseudo-label generator since no real classroom outcome data exists, and (b) comparison baseline.
- **Task design**, grounded in reading-comprehension research (not just generic learning analytics):
    - **Year 3–6 scope** — justified by Chall's stage model of reading development and the well-documented "fourth-grade slump," where comprehension difficulty first becomes distinguishable from decoding difficulty.
    - **Literal vs. inferential question tagging** — added because fluency/literal comprehension and inferential comprehension develop on different trajectories; blending them into one accuracy score hides this.
    - **Postdictive per-question judgment** ("do you think you got that right?") replacing a single upfront confidence rating — matches the exact method used in a study of 3rd–5th graders linking poorer metacognitive monitoring to reading comprehension difficulties. Produces a **bias score** and **accurate estimation count**.
    - **Pre-attempt hint tracking** — hints requested *before* any attempt are a stronger struggle signal than raw hint counts, per multi-semester K-12 research on unproductive hint use.
    - **Repair-attempted signal —** captures whether a hint or difficulty signal is followed by an actual repair action (reread/revision), grounded in comprehension-monitoring theory (detect → repair → recheck).
    - **Revisions treated directionally, not just counted —** a meta-analysis found most answer changes are incorrect→correct, so high revision counts alone shouldn't be flagged as a concern.
    - **Motivation grounding**: a study found teachers' judgements of reading difficulty aligned more with spelling results than actual reading assessment — direct evidence that even experienced educators misjudge this without structured support.
- **Implementation**: Next.js/TypeScript; features now captured include timing, hints (with pre-attempt flag), confidence + postdictive judgment, revisions (with direction), revisits, and literal/inferential accuracy split.
- **Model**: Random Forest trained on rule-engine pseudo-labels (no real outcome data available); explainability via feature importance.

2) Simulated Classroom Evaluation

Why selected: No real classroom outcome data exists at this stage (ethics/timeline), so there's no ground truth to evaluate against in the real world. A simulated classroom lets you generate data where the "correct answer" (which persona each learner represents) is known, so you can compute actual accuracy/precision/recall — something a handful of manual examples can't give you.

Is there similar research? Yes — this is an established method, not something improvised for this project. "Simulated learners" is a genuine ~30-year subfield of AI-in-Education research:

- The AIED Society's founder, John Self, described these as **"executable student models"** used as **"wind tunnels"** for testing tutoring-system designs before real deployment.
- Named systems in this tradition: **SimStudent, Cascade, Sierra, TutorGym**.

- A recent review, *"Evaluation Methods for Intelligent Tutoring Systems Revisited,"* explicitly lists **simulated learners** alongside learning curves and kappa scores as standard, recognised ITS evaluation techniques.
- Recent work even uses the term **"synthetic classroom cohorts"** for LLM-generated learner populations — conceptually very close to what you're doing.
- Worth citing honestly too: a systematic review of simulated learners proposes a "Turing-like test" for validating whether simulated learners actually resemble real ones — a good citation for your Limitations section, since your personas are self-defined, not validated against real learner distributions.

Tab 2

Planning / Notes:

Gaps in Current Tools / Systems:

- **a facilitator running a small reading group can see whether an answer was right or wrong, but not why it went wrong or right — and they have no time, mid-session, to diagnose that for every learner individually.** Existing tools (Kami, generic LMS dashboards) surface *outcomes* (score, completion) but don't interpret *behaviour* into something pedagogically actionable. That's the specific hole your project sits in — not "no digital reading tools exist," but "no tool turns interaction data into evidence-linked pedagogical judgement for the facilitator."

Contributions:

(a) A design contribution — translating reading-comprehension research into measurable system features. Most systems that do this kind of thing use generic behavioural proxies (time-on-task, click counts). Yours specifically operationalises literacy-research constructs: postdictive metacognitive judgement, pre- vs. post-attempt hint timing, literal/inferential/critical-analysis separation aligned to the NZ Curriculum's own reading sub-strands. That translation from literature into implementable, measurable features is itself a non-trivial piece of design work worth stating as a contribution on its own.

(b) An evaluation-methodology contribution — you had no access to a real classroom, which could read as a limitation. Reframe it: you adapted a ~30-year-old AIED technique (simulated learners, precedented by Self, TutorGym/SimStudent, Greer & Mark) to this specific domain, and then went a step further than most applications of it by adding the oracle-ceiling comparison — a way to quantify *how much headroom* exists before ever needing real deployment data. That's a reusable evaluation blueprint, not just a workaround. Anyone building a similar tool without classroom access could pick up your pipeline directly.

(c) An artifact contribution — the working prototype itself. Not a mockup, not a proposal: a functioning rule engine, dashboard, and evaluation pipeline someone could actually pilot. This is what makes "starting point for something similar" a defensible claim rather than a hedge — there's something concrete to start from.

3. How your findings back each one up

This is the part that turns "I built X and tested it" into "I built X, tested it, and here's what the evidence shows":

- **For (a)**: feature-importance results show the literature-grounded signal you deliberately added — confidence/accuracy miscalibration — dominates the model at 41%, well ahead of generic signals like response time (3%). That's direct evidence the research-grounded design choices, not the obvious surface metrics, are what carries predictive weight. State this explicitly: *"this validates prioritising literacy-specific behavioural constructs over generic interaction metrics."*
- **For (b)**: the rule engine reaches 70.6% precision / 44.6% recall; the oracle ceiling shows 95.5% precision / 84.3% recall is achievable on the *same features* with real outcome labels. That gap is your most important number in the whole project — it's quantified evidence of exactly how much value is currently left on the table, and it's the single clearest "future work" pointer you could hand to whoever picks this up next.
- **For (c)**: the dashboard, class-pattern detection, and personalised-baseline comparison are evidence the system goes beyond a fixed-threshold flag list toward something a facilitator could plausibly trust and use, not just a tech demo.

"This project does not claim to have solved the problem of facilitator capacity in small-group reading support. Instead, it contributes a research-grounded feature set, a working prototype built on it, and a reusable evaluation methodology that together establish a concrete, measurable foundation. The oracle-ceiling analysis in particular quantifies the specific direction future work should take — collecting real outcome labels to close the gap between the deployed model (44.6% recall) and the demonstrated ceiling (84.3% recall) — rather than leaving 'more research is needed' as a vague closing gesture."

Tab 3